

Optimizing Urban Delivery Routes Using Genetic Algorithms: A Smart City Vehicle Routing Approach

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Abstract— This research presents a Genetic Algorithm (GA) framework for optimizing urban delivery routes in smart-city logistics. It builds on the Capacitated Vehicle Routing Problem with Time Windows (CVRPTW). A synthetic dataset from CVRPLIB was expanded with realistic features, including service times, delivery time windows, and traffic factors to mimic real-world conditions. The EGA uses adaptive mutation and 2-Opt local search to improve route efficiency and stability. In comparative tests with a baseline GA, significant performance improvements were observed. The best fitness score rose from 2957.68 to 2219.44, and total travel distance dropped by 32.7% (from 1,769.55 to 1,190.96), while achieving 100% on-time deliveries. These results show the effectiveness of hybrid metaheuristics for sustainable and traffic-aware last-mile delivery planning in smart cities.

Keywords— *Vehicle Routing Problem (VRP); Smart City Logistics; Genetic Algorithm (GA); Time Windows; Route Optimization; Metaheuristics; Last-Mile Delivery.*

I. INTRODUCTION

The rapid growth of e-commerce and on-demand services has intensified pressure on urban logistics to reduce travel distance, energy consumption, and emissions while maintaining service reliability. This rise in delivery demand has changed how goods are distributed in cities and created complex challenges in last-mile logistics and transportation planning. Today, about 55% of the world's population lives in cities, and this number could reach 68% by 2050, according to the World Health Organization (WHO) [1]. As a result, effective urban transportation planning and logistics optimization are crucial for sustainable city growth.

To address these logistical and environmental challenges, cities worldwide are adopting smart-city frameworks. These frameworks use technology to improve mobility, sustainability, and resource management. Smart cities blend information and communication technologies (ICT), the Internet of Things (IoT), and real-time data analysis to enhance decision-making in transportation and logistics [2]. These advancements make smart cities not just technology-focused but also data-driven systems that support efficient, low-carbon urban logistics. They pave the way for better optimization methods like the Vehicle Routing Problem (VRP).

At the core of these challenges is the Vehicle Routing Problem (VRP). This problem involves creating cost-efficient routes for a fleet serving customers in different areas from one or more depots while adhering to operational limits. Since Dantzig and Ramser first introduced it, the VRP has become a major topic in operations research and transportation science [3]. The VRP is an NP-hard combinatorial optimization

problem, meaning its complexity increases rapidly as the problem size grows. Over the past sixty years, many variations of the VRP have appeared, including the Capacitated VRP (CVRP), VRP with Time Windows (VRPTW), and versions involving diverse or multi-depot fleets [4]. Efficient vehicle routing has long been important for urban logistics optimization because it helps reduce operational costs and boost delivery performance.

Genetic Algorithms (GAs), inspired by the process of natural selection, have become a powerful tool for solving complex combinatorial optimization problems by evolving a population of candidate solutions through selection, crossover, and mutation operations. In the case of the Vehicle Routing Problem (VRP), which is NP-hard and involves creating optimal routes with constraints like capacity and delivery needs, GA is effective because they balance global exploration with local improvement and can manage larger problem sizes [4]. Integrating GAs into VRP modelling offers a strong framework for vehicle routing in urban logistics and smart-city settings.

Despite these developments, there is still limited research that combines traffic congestion, service times, and delivery time windows into a single optimization framework. This study addresses that gap by developing a Genetic Algorithm (GA) for Smart-City Delivery Optimization. This GA minimizes a total weighted time goal that includes travel, service, and delay penalties. The algorithm is tested on a modified CVRPLIB dataset that considers congestion and time-window features, which reflect actual urban delivery conditions. The remainder of this paper is organized as follows: Section II reviews related literature on GA-based VRP optimization; Section III describes the proposed methodology and GA design; Section IV presents the results and discussion; Section V describes limitations and future works; and Section VI concludes the study.

II. LITERATURE REVIEW

Urban logistics, which involves managing goods delivery in city environments, is essential for smart city planning. Rapid urbanization has caused increased congestion, higher delivery demand, and environmental issues like air pollution [5]. Transportation in cities significantly contributes to CO2 emissions and traffic delays. This has led city authorities to seek smart mobility solutions. A systematic review by Santinha et al. found that routing logistic fleets for last-mile deliveries and urban waste collection are among the most examined smart city logistics applications, highlighting their importance for urban well-being [6]. Smart cities are

increasingly using tools like Internet of Things (IoT) sensors and artificial intelligence to improve delivery routes, reduce congestion, and lessen environmental harm. Effective urban logistics not only attracts business investments but also improves liveability by cutting traffic and emissions [5]. In summary, the smart city framework requires smart routing solutions to achieve efficiency and sustainability in urban goods distribution.

The Vehicle Routing Problem (VRP) is a foundational model for optimizing distribution routes. It involves creating the best routes for a group of vehicles to reach several customers from one or more depots, while considering different limits. In simple terms, it extends the Traveling Salesman Problem and is NP-hard, meaning exact optimization becomes difficult for large instances [7], [8]. Over decades of research, various VRP variants have been developed to address real-world needs [8]. Key variants include:

- i. Capacitated VRP (CVRP): vehicles have limited carrying capacity for goods.
- ii. VRP with Time Windows (VRPTW): each customer must be serviced within a specified time window.
- iii. Time-Dependent VRP: travel times vary by time of day (to model traffic congestion) [9]. For example, a route optimal at midnight may be suboptimal during rush hour.

These variants address the “*quality of service*” factors and operational constraints of urban logistics. As the VRP and its extensions are NP-hard, exact algorithms struggle on realistic city-sized problems. This has led to widespread use of heuristic and metaheuristic methods (tabu search, ant colony, genetic algorithms, etc.) to find near-optimal routes in reasonable time [8]. In summary, VRP variants offer a valuable set of tools to model urban delivery problems and solving them efficiently continues to be a major challenge for smart city logistics.

Genetic Algorithms (GAs) are a type of evolutionary metaheuristic inspired by natural selection. They maintain a group of candidate routes, or solutions, represented as chromosomes. GAs improves these solutions through processes like biological evolution: selection, crossover (recombination of routes), and mutation (random changes) [10]. They work well for complicated optimization problems like the Vehicle Routing Problem (VRP) [8]. The ability of GAs to explore a large solution space in parallel makes them effective for the complex search space of the VRP [8]. Prins (2004) [11] introduced a straightforward but effective hybrid genetic algorithm for the VRP that competes with advanced tabu search methods. This GA uses TSP-like permutation chromosomes without trip delimiters and includes a splitting process to optimally divide routes. This avoids infeasible solutions and eliminates the need for repair methods. It also includes a local search as a mutation operator and keeps a small, diverse population through spacing control to prevent early convergence. When tested on the Christofides and Golden benchmark datasets, the algorithm performed better than most tabu search and simulated annealing methods in terms of solution quality and stability, achieving nearly optimal results on large cases. This shows that GAs, despite their simplicity, can deliver high-quality solutions that compete with specialized methods.

However, standard genetic algorithms face challenges when applied to complex vehicle routing problems. Premature convergence can occur, resulting in a loss of diversity in the population and making it stuck in suboptimal patterns [12]. Ding et al. (2024) proposed an Improved Dynamic Crossover and Mutation Rate Genetic Algorithm (IDCMGA) to optimize urban travel routes between Beijing Union University and Beijing South Railway Station. The study showed that adjusting crossover and mutation rates in IDCMGA improved global search efficiency, convergence speed, and solution stability. Experimental results demonstrated a 15-60% reduction in travel cost variance and a significant improvement in route optimization [12]. This approach helps with smart city transportation planning by providing an efficient and flexible method for real-time urban route optimization.

Similarly, Wang et al. (2020) [13] combined the search capabilities of genetic algorithms with column generation (GA-CG) to plan electric bus routes more efficiently. When tested with real data from Qingdao, China, the GA-CG model performed about 40 times faster than traditional methods and used fewer vehicles while still covering almost all routes. These examples show that, with careful design, genetic algorithm methods can effectively solve vehicle routing problems that are otherwise very challenging, making them appealing for urban route optimization.

Genetic algorithms are used in many urban delivery and smart mobility situations, often leading to significant improvements in operations. One key application is dynamic urban delivery routing, which adjusts vehicle routes based on traffic congestion and real-time events. Kumar and Panneerselvam (2017) showed how Genetic Algorithms can optimize urban delivery routes, especially under changing traffic conditions modelled as the Time-Dependent VRP with Time Windows (TDVRP-TW) [9]. Their method demonstrated how GA-based models can effectively respond to varying travel speeds and urban congestion, making them ideal for real-time, last-mile delivery in smart city settings.

Another practical example comes from urban freight distribution in Quito. Herrera-Granda et al. (2023) proposed a hybrid Genetic Algorithm (GA) with the Nearest Neighbour (NN) heuristic to enhance urban delivery routing for an auto parts company. The model was based on the Capacitated VRP with Time Windows (CVRPTW) and considered real traffic, vehicle limits, and delivery schedules. Their approach cut total travel distance by nearly 19% compared to manual planning. This shows that GA-based hybrid models can effectively improve urban logistics and last-mile delivery performance [14].

GA-based methods have also been modified to handle the uncertainties and complexities of last-mile delivery in smart cities. Tang (2025) introduced a Fuzzy Genetic Algorithm (FGA) for urban terminal deliveries. This algorithm uses fuzzy logic to manage uncertainties like traffic jams, weather, and changing customer demands. By optimizing several goals—delivery time, transportation cost, and customer satisfaction—the FGA creates more reliable and efficient routes in complex urban settings. Experimental results indicate significant performance improvements ($p < 0.05$) in reducing travel time and cost while enhancing service quality [15]. This hybrid method serves as a strong and flexible solution for modern smart city logistics, supporting both operational efficiency and sustainability. In summary, various

studies have shown that GA methodologies can be effectively applied to urban logistics, providing real benefits such as shorter routes, lower fuel consumption, and better adherence to delivery schedules in the smart city context.

Optimizing delivery routes is closely tied to sustainability and green logistics in cities. Efficient routes allow vehicles to drive fewer miles and spend less time idling. This directly reduces fuel use and emissions. Li et al. (2025) proposed an Improved Non-Dominated Sorting Genetic Algorithm III (INSGA-III) to address the Green Vehicle Routing Problem with Time Windows (GVRPTW). Their focus was on lowering delivery costs and carbon emissions while increasing customer satisfaction. When tested on benchmark and real-world urban logistics data, this model showed lower fuel consumption and emissions compared to other algorithms, promoting energy-efficient and sustainable delivery planning [16]. This study highlights the importance of genetic algorithms in promoting green and low-carbon urban logistics.

Although significant progress has been made in using genetic algorithms for vehicle routing, several research gaps still exist, especially in smart city urban delivery systems. Studies like Ding et al. [12] and Kumar & Panneerselvam [9] have included dynamic travel times, but few have integrated these with capacity, time-window, and traffic-aware constraints in a single framework suitable for multi-vehicle last-mile operations. Additionally, most previous works focus on minimizing total distance as the only goal, ignoring the importance of service quality and workload balance among vehicles. These factors are essential for sustainable urban logistics [16].

This research addresses these gaps by applying a traffic- and time-aware Genetic Algorithm for the Capacitated VRP with Time Windows (CVRPTW). It uses dynamic mutation rates and embedded 2-Opt local improvement to allow for flexible exploration and refinement of candidate routes. The algorithm's performance is assessed using various Key Performance Indicators (KPIs), including Total Distance, Total Travel plus Service Time, On-Time Delivery Rate, and Route Balance Index. This provides a well-rounded evaluation of efficiency, punctuality, and sustainability.

In summary, this study offers a tested and reproducible framework that puts into practice the theoretical advances found in earlier literature. Future work may build on this foundation to develop hybrid GA-ML models for real-time routing and connect with live IoT traffic data to create fully adaptable urban delivery systems.

III. METHODOLOGY

A. Problem Formulation

The delivery routing problem in this research is modelled as a Capacitated Vehicle Routing Problem with Time Windows (CVRPTW), which is a practical extension of the standard Vehicle Routing Problem (VRP). The goal is to plan efficient routes for a fleet of vehicles that start and end at a central depot while delivering goods to all customer locations. Each vehicle has a limited carrying capacity, and every customer must be served exactly once within a specific delivery time window.

In this study, each customer node has three main attributes: demand, which indicates the quantity to be delivered; service

time, which is the time taken to unload or finish the delivery; and a time window, defined by an earliest (Ready) and latest (Due) delivery time. These constraints make the problem more representative of real-world logistics, where vehicles must meet delivery schedules and capacity limits at the same time. To make the routing model more realistic for a smart city, traffic conditions are also considered. Each location in the dataset has a Traffic Factor between 0.6 and 1.2, indicating the level of congestion in that area. Lower values mean slower movement due to traffic, while higher values reflect smoother flow.

B. Overview of Approach

The main goal of the system is to reduce total travel time and operational costs while ensuring all deliveries happen within their scheduled time frames. The approach uses evolutionary search and traffic-aware modelling to create efficient, practical, and sustainable routing plans for last-mile delivery operations. The entire workflow has five main stages: (1) preparing a smart dataset sourced from the CVRPLIB benchmark, which includes urban details like traffic multipliers and delivery windows; (2) using a GA-based optimizer to create feasible delivery routes; (3) visualizing and analysing optimized solutions; and (4) evaluating performance based on Key Performance Indicators (KPIs). Two algorithm configurations were created for comparison: a Baseline GA that represents a standard implementation with fixed parameters and distance-based optimization, and an Advanced GA that includes adaptive mutation control, 2-Opt local route refinement, and traffic-aware travel-time calculation. The comparison between these two models shows how effective hybrid and adaptive strategies can be in improving route efficiency and service reliability in real smart-city scenarios.

C. Data Preparation and Smart City Features

A benchmark instance from the CVRPLIB dataset was chosen as the foundation for this study. This dataset outlines the customer locations, their demands, and the depot position for a standard vehicle routing scenario with capacity limits. To mimic realistic delivery conditions in a smart city, more attributes were created to reflect urban factors that affect delivery performance. These include Service Time, Delivery Time Windows, and Traffic Factors.

Each customer node received a Service Time between 2 and 7 minutes, representing the unloading or handover time at the delivery point. Time Windows were created to reflect scheduling limits: the earliest possible delivery time (Ready) ranged from 60 to 360 minutes after the start of the day, and each window lasted from 240 to 420 minutes to determine the latest allowable delivery time (Due). This approach ensured that each customer had a unique delivery schedule.

An important addition to the dataset is the Traffic Factor, which shows the level of road congestion near each delivery location. Each node was assigned a Traffic Factor between 0.6 and 1.2. Lower values indicate slower movement due to congestion, such as in busy downtown areas, and higher values indicate smoother traffic flow, like in suburban or highway regions. The travel time between two points i and j was computed as the Euclidean distance divided by the average of their traffic factors as shown in (1):

$$T_{ij} = \frac{D_{ij}}{(TrafficFactor_i + TrafficFactor_j)/2} \quad (1)$$

where D_{ij} is the Euclidean distance between nodes i and j .

The resulting dataset keeps the original CVRPLIB structure and includes these smart-city variables. This dataset offers a realistic foundation for assessing both the baseline and advanced GA optimization models in situations that resemble real-world urban delivery operations. Table 1 provides a summary of the dataset.

Table 1: A detailed summary of the dataset and modification

Features	Description	Range/Source
X, Y	Node Coordinates	CVRPLIB benchmark
Demand	Customer Order Quantity	CVRPLIB benchmark
Service Time	Stop Service duration	2-7 min
Ready / Due	Time Window (min)	60-780
Traffic Factor	Congestion multiplier	0.6 – 1.2

D. Genetic Algorithm Design

The optimization process in this study relies on a Genetic Algorithm (GA), a population-based method inspired by natural selection. The GA evolves a set of potential delivery routes through steps like biological evolution: selection, crossover, mutation, and elitism. Everyone in the population represents a complete sequence for visiting customer nodes. The algorithm aims to find the most efficient routes that minimize total travel time while adhering to delivery time windows and vehicle capacity limits.

In each generation, every solution is assessed using a fitness function that factors in total travel time, service durations, waiting time, and lateness penalties. Solutions with lower total costs have a higher chance of being selected for reproduction. Tournament selection with a group size of three is used to pick parent solutions. The Order Crossover (OX) operator combines the route sequences of two parents while keeping the relative order of customer visits. The Swap Mutation operator randomly exchanges two customers in a route, adding variability and preventing early convergence. Elitism ensures that the two best solutions from each generation transfer directly to the next, maintaining the best performance achieved so far.

Two variations of the GA were set up for comparison. The Baseline GA is a standard version where the mutation rate is fixed at 0.2, and no local search is applied. This version focuses solely on exploring route combinations through genetic operators and serves as a reference for measuring improvements. In contrast, the Advanced GA introduces several enhancements aimed at improving speed and quality of solutions. First, the mutation rate is adjusted over generations, starting at 0.2 and decreasing to zero, allowing for more exploration in early stages and more exploitation later on. Second, a 2-Opt local search operator is used on each vehicle's route to eliminate unnecessary loops and cut total travel time. This process checks pairs of edges in a route and swaps them if it creates a shorter or faster path. Finally, the Advanced GA utilizes traffic-aware travel times from the Traffic Factor values, ensuring that optimization choices consider real congestion rather than just geometric distances.

E. Performance Evaluation

The performance of both the Baseline GA and Advanced GA was assessed using several Key Performance Indicators (KPIs) that are commonly used in vehicle routing and smart-city logistics. These KPIs were calculated after the optimization process with a special evaluation function. This function simulates vehicle travel, service, and waiting behavior based on the optimized routes. The metrics give a detailed view of both operational efficiency and service reliability. The main KPIs used in this study are as follows:

- Total Distance:** The total Euclidean distance covered by all vehicles across all routes.
- Total Travel + Service Time:** The total time spent traveling and servicing customers, reflecting overall operational efficiency.
- Total Waiting Time:** The total waiting time across all deliveries, representing the time spent idle when a vehicle arrives before the customer is ready.
- Total Lateness:** The total delay accumulated when deliveries exceed their scheduled times.
- On-Time Delivery Rate:** The percentage of deliveries completed within their time windows, measuring service punctuality.

IV. RESULTS AND DISCUSSION

The Enhanced Genetic Algorithm (EGA) was tested against the Baseline GA to evaluate its ability to optimize urban delivery routes while considering time-window and capacity limits. Both algorithms used the same dataset (A-n37-k6-smart), with a population size of 100 and 1200 generations. The comparison focused on convergence behavior, route efficiency, and service reliability, using defined Key Performance Indicators (KPIs).

The comparative results between the Baseline GA and the Enhanced GA as shown in table 2 reveal a clear improvement in route efficiency and optimization performance. The Enhanced GA achieved a final fitness score of 2219.44, which is about 25% better than the Baseline GA's score of 2957.68, confirming that the enhancements significantly boosted convergence. The integration of adaptive mutation and 2-Opt local search in the Enhanced GA allowed it to continually refine solutions across generations. This approach helped avoid premature stagnation and encouraged a balanced exploration and exploitation of the search space.

Table 2: KPI Comparison Between Baseline and Enhanced Genetic Algorithm

Metrics	Baseline GA	Enhanced GA
Total Distance	1,769.55	1,190.96
Total Travel + Service Time	2,220.02	1,469.91
Total Waiting Time	737.66	749.53
Average Waiting per Stop	20.49	20.82
Total Lateness	0.00	0.00
On-Time Delivery Rate	100.00	100.00

From a performance perspective, the total distance and total travel plus service time dropped by 32.7% and 33.8%, respectively. This large reduction shows that the Enhanced GA produced shorter, more efficient routes, reducing unnecessary travel while utilizing the 2-Opt optimization step to remove less effective route segments. The slight rise in average waiting time per stop (AWT) from 20.49 to 20.82 minutes suggests that, although the Enhanced GA focused on reducing total operational time, some vehicles arrived a bit earlier at customer locations due to tightened route sequencing. However, this trade-off did not harm delivery reliability—all deliveries were completed on time (OTD% = 100%), with no lateness reported in either model.

The convergence behavior also supports these results. As illustrated in Fig. 1, the Baseline GA plateaued after about 400 generations, while the Enhanced GA, shown in Fig. 2, kept improving until around 1100 generations, resulting in smoother and lower-cost convergence. This indicates that the adaptive mutation rate effectively sustained population diversity early on, while 2-Opt local refinement consistently enhanced offspring solutions in later generations. Consequently, the Enhanced GA achieved better overall stability and efficiency, producing routes that were shorter, more balanced, and fully compliant with delivery constraints.

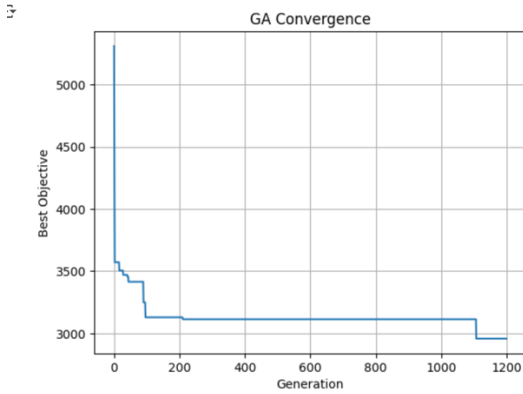


Figure 1: Baseline GA Showing Early Stagnation

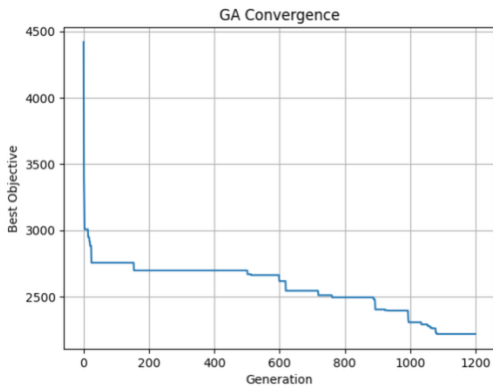


Figure 2: Enhanced GA Convergence Showing Faster and Smoother Improvement

Fig. 3 and Fig. 4 show the delivery routes created by the Baseline and Enhanced Genetic Algorithms. The Enhanced GA produces smoother, more compact, and less overlapping routes than the baseline. This indicates better spatial grouping and route efficiency.

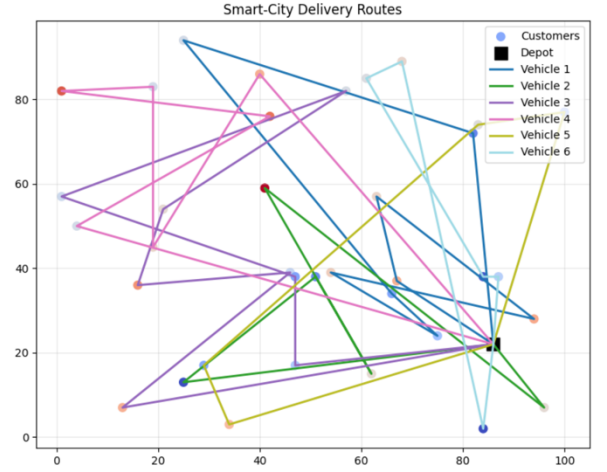


Figure 3: Delivery Routes Using the Baseline GA, Showing Overlapping Paths.

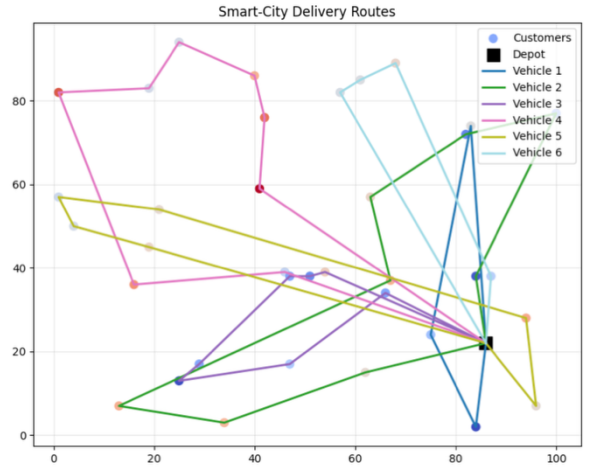


Figure 4: Delivery Routes Using the Enhanced GA, Showing Smoother Paths.

In the context of smart-city logistics, these findings suggest that using evolutionary search can significantly boost delivery operations. The reductions in total distance directly led to lower fuel consumption, which are essential for sustainable urban transportation systems.

V. LIMITATIONS AND FUTURE WORKS

While the proposed Genetic Algorithm (GA) achieved significant improvements in route efficiency and convergence stability, some limitations remain. The study used a synthetic smart-city dataset from CVRPLIB, which, although is realistic, but might not fully represent the unpredictable nature of real-world urban traffic, changing demand patterns, or unexpected delivery delays. Moreover, the algorithm assumes that traffic factors and time windows remain fixed during the simulation. In real smart-city logistics, traffic conditions and delivery priorities change constantly. This requires real-time re-optimization and adaptive control mechanisms.

Future research can address these challenges by adding real-time traffic sensing and dynamic re-routing. This would allow the algorithm to adjust to changing urban conditions. Additionally, extending the framework to multi-objective

optimization would be useful. Balancing travel efficiency, fuel use, and environmental impact would make it more relevant for sustainable urban logistics. Finally, scaling the model to larger datasets and exploring parallel or hybrid GA-based methods could further improve computational efficiency.

VI. CONCLUSION

This research presented a framework for a Genetic Algorithm (GA) to optimize urban delivery routes under smart-city constraints. It built on the capacitated Vehicle Routing Problem with Time Windows (VRPTW) by adding features like service times, dynamic traffic factors, and customer time windows. This created a more realistic delivery environment. Comparative analysis between with Baseline GA demonstrated that incorporating adaptive mutation rates and 2-Opt local search significantly improved solution quality.

The Enhanced GA achieved a 32.7% reduction in total distance (from 1,769.55 to 1,190.96) and improved the best fitness score from 2957.68 to 2219.44, while keeping 100% on-time delivery performance. These results demonstrate the potential of GA-based methods for effective and sustainable last-mile logistics in smart cities.

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